

Arya Shrestha

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EDUCATION

Marist University

Bachelor of Science in Computer Science **CGPA: 3.9**

Poughkeepsie, NY

Expected May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Software Development I & II, Data Analysis and Visualization, Database Management, Discrete Mathematics, Data Communications

EXPERIENCE

Software Engineer Intern

September 2025 – February 2026

Legacy Louisiana

Remote

- Built a **multi-tenant legal SaaS** platform serving **60+** attorney subscribers, engineering estate planning workflows for Louisiana trust & succession laws with **PostgreSQL**, reducing document preparation time by **40%** through automated **unit** and **integration testing** pipelines.
- Developed **full-stack** infrastructure with **Express.js** and **PostgreSQL** handling **3+** tiers of attorney-client hierarchies and state-specific legal validation, supporting **15x** projected growth to **1,000+** practices.

Software Engineer Intern

June 2025 – September 2025

Docere

Remote

- Created an **AI-driven** learning platform for **100+** children with **Redis** caching, **SQL**-based progress tracking, and a **Retrieval-Augmented Generation (RAG)** layer that pulled lesson context and student history into each prompt, boosting AI response speed by **60%**.
- Deployed microservices using **Docker/Kubernetes**, integrated auth via **Auth0**, and implemented **CI/CD** pipelines while securing **\$10,000** in funding and launching pilots across **2 schools** and **1 library**.

PROJECTS

Claude Vessel | *Electron, React, TypeScript, node-pty, xterm.js*

March 2026

- Solved **context-switching overhead** for **Claude Code** developers by building a **multi-workspace terminal manager** supporting up to **6 concurrent sessions** with persistent state, tab organization, and drag-to-sidebar workflows using **Electron** and **React**.
- Eliminated **terminal output loss** during session navigation by engineering a **custom node-pty IPC layer** backed by a **100KB per-terminal ring buffer** that replays full session history in **sub-second** on remount.

Distributed Rate Limiter | *Node.js, Redis, Nginx, Docker, Supabase, D3.js*

January 2026

- Built a **full-stack SaaS rate limiting platform** with user authentication, **API key provisioning** tested across **100+ keys**, and per-key usage tracking by integrating **Redis**, **Nginx** load balancing, and **Supabase** for auth and data persistence.
- Surfaced **real-time API usage insights** for users by designing a **D3.js** dashboard that visualizes per-key request trends, hit counts, and rate limit events across a **Docker**-containerized microservices backend.

Posturize (HackPrinceton) | *Python, React JS, Flask, OpenCV, MediaPipe, D3.js*

November 2025

- Engineered a **real-time posture analysis** system entirely solo in a **36-hour** hackathon sprint by architecting a **Flask REST API** pipeline integrating **OpenCV** and **MediaPipe** to detect and classify **30+** body landmarks.
- Visualized live posture tracking trends for users by building a **D3.js** dashboard that surfaces historical posture data, giving users actionable feedback on long-term postural habits.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, Java, C#, SQL

Systems & Tools: Linux, Docker, Kubernetes, Nginx, Git & GitHub, AWS, Wireshark, Redis

Networking: TCP/IP, REST APIs, OAuth 2.0, network protocol fundamentals, Data Communications coursework

Databases: PostgreSQL, MySQL, Redis, NoSQL (MongoDB)

Additional Skills: Unit & Integration Testing, CI/CD Pipelines, Agile/Scrum, Automation Scripting, Debugging

LEADERSHIP & INVOLVEMENTS

HeadStart Fellowship Tech Fellow (Spring 2026), Propel2Excel Phase 2 Fellow - 1% acceptance (2025-Present), ColorStack Member - Resume reviews and peer mentorship for freshmen, Notion Campus Leader